

CPEN 416 / EECE 5710

Lecture 20: Correcting arbitrary single-qubit errors; intro to stabilizers

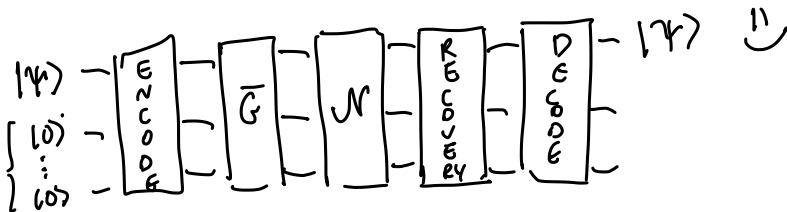
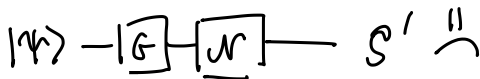
Thursday 20 November 2025

Announcements

- Quiz 9 on Tuesday
- Last tutorial assignment on Monday
- Reminder to fill out weekly surveys for project

Last time

We discussed the motivation for quantum error correction, and developed a mathematical (/graphical) model for noise.



Last time

We saw our first quantum error correcting code: the quantum repetition code, or bit flip code.

$$\mathcal{E}(|0\rangle) = |000\rangle = |0\rangle_L$$

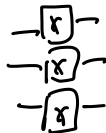
$$\mathcal{E}(|1\rangle) = |111\rangle = |1\rangle_L$$

$$\alpha|0\rangle + \beta|1\rangle \quad \begin{array}{c} \bullet \quad \bullet \\ | \quad | \\ \oplus \quad \oplus \\ | \quad | \\ \oplus \quad \oplus \end{array} \quad \rightarrow \quad \alpha|0\rangle_L + \beta|1\rangle_L$$

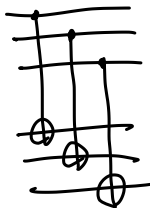
Last time

We designed logical operations for the bit flip code.

$$|0\rangle_L \leftrightarrow |11\rangle_L$$

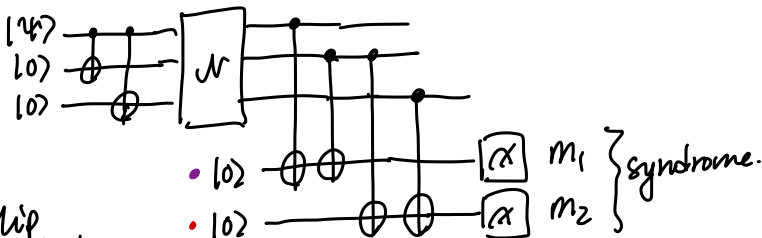


$CNOT_L$



Last time

We designed a circuit to detect if a bit flip error occurred, and identify its location.



e.g. bit flip
on qubit 1

$$\alpha |010\rangle + \beta |101\rangle$$

11 $\rightarrow$ bit flip on Q1

$$01 : \text{odd} \quad \left(\sum_i \text{bits} \right) \bmod 2$$

$$11 : \text{even}$$

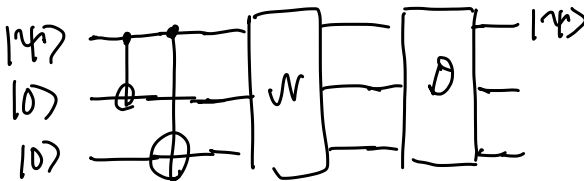
Learning outcomes

Today:

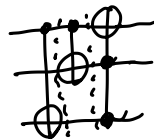
- ➊ Correct bit *and* phase flip errors with the 9-qubit code
- ➋ Outline the conditions under which errors can be corrected
- ➌ Define the stabilizers of a quantum error correcting code

Bit flip code: recovery operation

Design a circuit to *recover* the state of the first qubit



$$\begin{aligned}
 \alpha |000\rangle + \beta |111\rangle &\longrightarrow (\alpha |0\rangle + \beta |1\rangle) \otimes \text{stuff} \\
 \alpha |100\rangle + \beta |011\rangle &\longrightarrow (\alpha |0\rangle + \beta |1\rangle) \otimes \text{stuff} \\
 &\vdots \\
 \alpha |010\rangle + \beta |101\rangle & \\
 \alpha |001\rangle + \beta |110\rangle &
 \end{aligned}$$



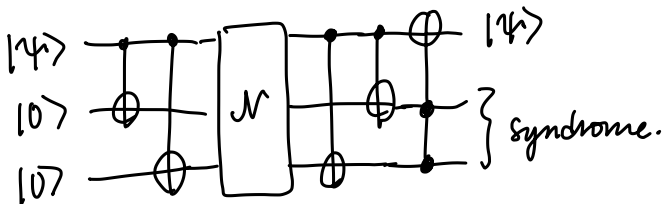
parity
qs 0 & 2

parity
qs 1 & 2



$$\begin{aligned}
 \alpha |010\rangle + \beta |110\rangle \\
 = (\alpha |0\rangle + \beta |1\rangle) |10\rangle
 \end{aligned}$$

Bit flip code: full circuit for bit flip code



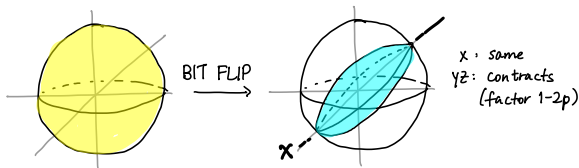
2 bit flips:

$$\alpha|011\rangle + \beta|100\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle$$

$$\alpha|0\rangle + \beta|1\rangle$$

Bit flip code: analysis

Imagine any possible state that can go through the channel:



Exercise: Determine $F(|\psi\rangle, \Phi(|\psi\rangle\langle\psi|))$ where Φ is the *bit flip channel* with parameter p .

$$\begin{aligned}
 F(|\psi\rangle, \Phi(|\psi\rangle\langle\psi|)) &= \langle\psi|\Phi(|\psi\rangle\langle\psi|)|\psi\rangle \\
 &= \langle\psi|((1-p)|\psi\rangle\langle\psi| + pX|\psi\rangle\langle\psi|X)|\psi\rangle \\
 &= (1-p)\underbrace{\langle\psi|\psi\rangle\langle\psi|\psi\rangle}_1 + p\cdot\underbrace{\langle\psi|X|\psi\rangle\langle\psi|X|\psi\rangle} \\
 \langle X \rangle^2 &= 1 : |+\rangle, |-\rangle \\
 \langle X \rangle^2 &= 0 : |0\rangle, |1\rangle \\
 &= 1 - p + p\langle X \rangle^2
 \end{aligned}$$

Bit flip code: analysis

Compare overall fidelity of the state *after the error*, vs. fidelity *after recovery*.

After the error

$$F(|\psi\rangle, \mathbb{E}(|\psi\rangle\langle\psi|)) = 1 - p + p \langle X \rangle^2 \geq 1 - p$$

After recovery, have the state

$$\mathcal{R}(\mathbb{E}(|\psi\rangle\langle\psi|)) = (1-p)^3 |\psi\rangle\langle\psi| + 3p(1-p)^2 |\psi\rangle\langle\psi| + \text{bit flip cases}$$

$$F(|\psi\rangle, \mathcal{R}(\mathbb{E}(|\psi\rangle\langle\psi|))) \geq (1-p)^3 + 3p(1-p)^2$$

Phase flip errors

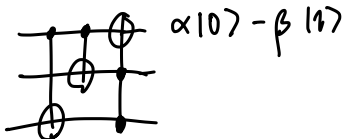
With our encoding

$$\alpha |000\rangle + \beta |111\rangle$$

↑

and appropriate circuitry, we can correct *bit flip errors* but not phase flip errors.

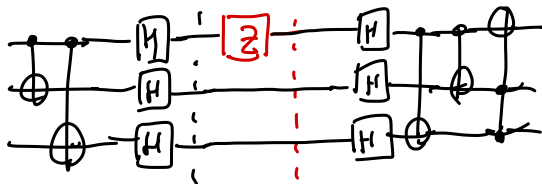
$$\alpha |000\rangle - \beta |111\rangle$$



Phase flip code: encoding circuit

Main idea: make phase flip errors “look like” bit flip errors.

$$\mathcal{E}(\alpha|0\rangle + \beta|1\rangle) = \alpha \underbrace{|+++ \rangle}_{|0\rangle_L} + \beta \underbrace{|--- \rangle}_{|1\rangle_L}$$

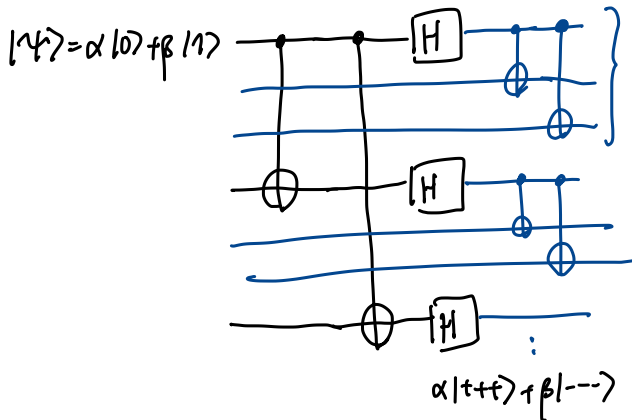


$$\alpha|+++ \rangle + \beta|--- \rangle$$

$$\alpha| - + + \rangle + \beta| + - - \rangle$$

Shor code

To correct a combination of one bit flip and/or phase flip error, we can *concatenate* codes.



Shor code

The Shor code can correct one *arbitrary* error on a *single* qubit.

$$|0\rangle_L : |+++ \rangle + |--+ \rangle$$

(phase flip)

$$|0\rangle_L = \frac{(|1000\rangle + |1111\rangle)(|1000\rangle + |1111\rangle)(|1000\rangle + |1111\rangle)}{2\sqrt{2}}$$

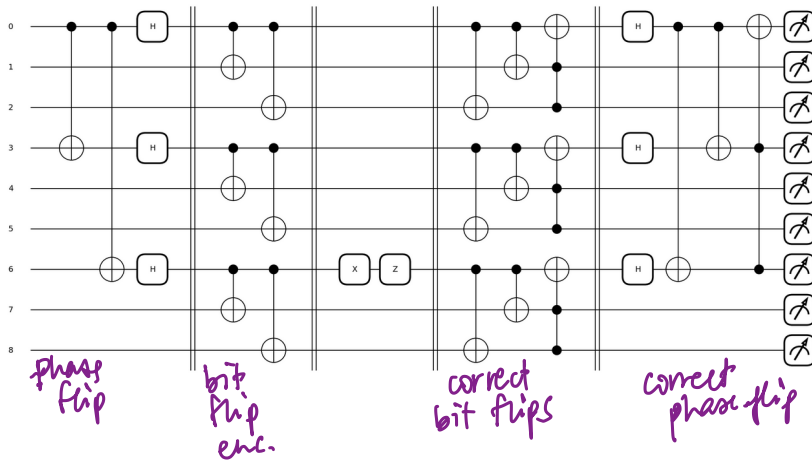
$$|1\rangle_L = \frac{(|1000\rangle - |1111\rangle)(\quad)(\quad)}{2\sqrt{2}}$$

Imagine a bit + phase flip on qubit 6:

$$\frac{(|1000\rangle + |1111\rangle)(|1000\rangle + |1111\rangle)(-|100\rangle + |1011\rangle)}{\sqrt{2}}$$

correct
bit flip

Shor code



Next time

Next class:

- More stabilizer codes: phase flip, $[[5, 1, 3]]$ code, Steane code

Action items:

- 1 Quiz 9
- 2 Work on project

Recommended reading:

- From this class: Codebook EC; 10.3, 10.5
- For next class: Codebook EC; 10.1-10.5